

Muthukumar

Software Development Engineer · Golang · System Design · Cloud · Full SDLC

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PROFESSIONAL SUMMARY

Software Development Engineer with strong foundations in **system design, distributed systems**, and the full SDLC — from architecture and low-level design through deployment and monitoring. Primary expertise in **Golang** backend engineering, with breadth across React, Python, and Node.js. I build and own production-grade cloud-native infrastructure on AWS with Terraform and Docker, driving automated CI/CD. Strong in DSA and performance optimisation, with an Agile mindset and a track record of shipping robust software in fast-paced environments. Have collaborated directly with **Founding Engineers** on architecture and delivery of multiple products.

CORE SKILLS

- Languages:** Golang (primary), Python, Node.js, JavaScript (ES6+), PHP
- Backend Engineering:** REST APIs, gRPC, Protocol Buffers, SSE, Microservices, API-first design, JWT, Middleware design
- Frontend:** React, TypeScript, Tailwind CSS, Redux-Saga, HTML, CSS/SCSS, Bootstrap
- Cloud & Infrastructure:** AWS (EC2, S3, RDS, EKS, Lambda, VPC, CloudFront, Route 53, IAM, SES, Secrets Manager, API Gateway), Terraform, Docker, Docker Compose
- Databases:** PostgreSQL, MongoDB, Redis, MySQL
- CI/CD & Git:** GitHub Actions, GitLab CI/CD, Git hooks, trunk-based & feature branching, semantic versioning
- Architecture & Design:** Microservices, Monolith, LLD, SOLID, DRY, 12-Factor, System Design, code reviews
- Monitoring:** CloudWatch, Prometheus, structured logging, performance profiling
- ML (Applied):** scikit-learn, pandas, OpenCV — KNN, Decision Trees, AdaBoost
- DSA:** Trees, graphs, tries, heaps, DP, DFS/BFS, sliding window, two pointers, backtracking
- Agile:** Scrum, sprint planning, cross-functional collaboration, requirement analysis

WORK EXPERIENCE

Company	Role	Duration	Mode
Nelsonin Research Pvt Ltd	Junior React Developer – Intern	Aug '22 – Nov '22	Remote
NRG Phoenix Technology	Full Stack Developer	Jul '23 – Jun '24	On-Site
Nelsonin Research LLC	Software Development Engineer	Jun '24 – Present	Remote

PROJECTS

Clenz – AI-Powered Skincare Ecosystem (Mobile App)

Jul 2024 – Present · Backend & Cloud Lead, ML Integrator · *Worked directly with the Founding Engineer*
Production mobile app delivering personalized skincare routines and real-time AI face analysis. Owned the full Golang backend and AWS infrastructure end-to-end alongside the Founding Engineer — from microservice design to deployment.

- Golang (Gin) microservices: auth, product, ML inference, community, and routine services with SSE-based real-time face scan streaming
- ML pipeline: scikit-learn (KNN, Decision Trees, AdaBoost) + OpenCV for skin concern detection; OCR product input and duplicate finder
- Hybrid storage: PostgreSQL (relational data), MongoDB (product catalogue), Redis (caching/sessions), Apache Solr (search)
- Full AWS setup: EC2, RDS, S3, IAM, Secrets Manager — provisioned with Terraform; CI/CD via GitHub Actions; observability with Prometheus

Stack: Golang (Gin) · REST · SSE · gRPC · scikit-learn · OpenCV · PostgreSQL · MongoDB · Redis · Solr · Flutter · AWS · Docker · Terraform · GitHub Actions · JWT · Prometheus

Clenz Web Application – Internal AI Skincare Dashboard

Jul 2024 – Present · Full Stack Lead, Cloud · *Worked directly with the Founding Engineer*

Web application module — a companion to the Clenz mobile app. Led full-stack development from React frontend to Golang backend services and managed all AWS infrastructure.

- React + TypeScript + Tailwind CSS frontend: AI face scan analyser with real-time SSE streaming, analytics dashboard, compatibility scoring, and product comparison engine
- Golang backend: dedicated auth service + shared Clenz app server services (product, routine, community, blog) with clean service boundaries
- PostgreSQL as primary database; Redis for session/cache; Apache Solr for product search
- AWS infrastructure (EC2, RDS, S3, CloudFront) fully managed and provisioned with Terraform; automated CI/CD via GitHub Actions

Stack: React · TypeScript · Tailwind CSS · Golang (Gin) · REST · SSE · PostgreSQL · Redis · Solr · AWS (EC2, RDS, S3, CloudFront) · Docker · Terraform · GitHub Actions · JWT · Prometheus

Catepilla – Child Onboarding Assessment & Dashboard

Full Stack Lead, Cloud · *Worked directly with the Founding Engineer*

Internal platform for doctors, therapists, parents, and admins to manage child onboarding assessments, track progress, and generate insights. Led full stack delivery and owned all cloud infrastructure from day one.

- React + TypeScript + Tailwind CSS frontend: child onboarding flows, assessment questionnaires, progress dashboards, and role-based views for doctors, therapists, parents, and admins
- Golang microservices: dedicated auth service (JWT, RBAC) and questionnaire service — clean separation of concerns with PostgreSQL as the single source of truth
- AWS infrastructure (EC2, RDS, S3) fully managed end-to-end; Terraform for IaC; GitHub Actions CI/CD pipeline

Stack: React · TypeScript · Tailwind CSS · Golang (Gin) · REST · PostgreSQL · AWS (EC2, RDS, S3) · Docker · Terraform · GitHub Actions · JWT

Origine Questionnaire – Predictive Health-Tech Platform

Jul 2024 – Present · Backend & Cloud Lead, ML Integrator · ai.originesecret.com

Health-tech web app assessing disease risk and recommending personalised wellness products via real-time ML predictions.

- FastAPI (Python) backend with scikit-learn models (KNN, Decision Tree, AdaBoost); Vite + TypeScript frontend
- AWS infrastructure: EC2, RDS (PostgreSQL), S3, CloudFront, SES — Docker Compose + GitHub Actions for deployments

Stack: FastAPI · scikit-learn · pandas · TypeScript · Vite · PostgreSQL (RDS) · AWS · Docker Compose · GitHub Actions

Formtheta – SaaS Survey & Form Builder · formtheta.com

Jul 2024 – Present · Full Stack & Cloud Lead

Scalable SaaS platform for building and analysing interactive forms with real-time data processing.

- React + Redux-Saga frontend; Golang REST API backend with hybrid storage (PostgreSQL, MongoDB, Redis)
- AWS (EC2, RDS, S3, VPC) provisioned with Docker; GitHub Actions CI/CD

Stack: React · JavaScript · Redux-Saga · Golang · PostgreSQL · MongoDB · Redis · AWS · Docker · GitHub Actions

Agas National – SaaS Admin Portal · portal.agasnational.com.au

Jul 2023 – Jun 2024 · Full Stack Developer

Centralised admin portal for managing bookings, service tracking, and employee operations in the gas services sector.

- Laravel/PHP monolith with Blade templates; jQuery + Bootstrap for responsive UI; REST API integrations; GitLab CI/CD

Stack: Laravel · PHP · Bootstrap · jQuery · REST APIs · GitLab CI/CD

Career Portal – Job & E-Learning Platform · careers.nrg-phoenix.com

Jul 2023 – Nov 2023 · Full Stack Developer

Job application and e-learning platform integrated with Moodle LMS for course delivery and progress tracking.

- Laravel + Blade end-to-end; Moodle REST API integration; GitLab CI/CD deployments

Stack: Laravel · PHP · Bootstrap · jQuery · Moodle (LMS) · REST APIs · GitLab CI/CD

PROBLEM SOLVING & DSA

Practical experience applying trees, graphs, tries, heaps, dynamic programming, and algorithmic patterns (DFS, BFS, sliding window, two pointers, backtracking) to solve complex problems with optimal time and space complexity.

EDUCATION

B.Sc Information Technology — MS University, Tamil Nadu

CERTIFICATIONS

